

Introduction to Contract Theory and Auctions

Bayesian Auctions

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- The theory of auctions is considered to be one of the **most successful applications of game theory**, in particular of games of incomplete information.
- We will model auctions as games with incomplete information, i.e., as a Bayesian game.
- The main interest is to study the Bayesian Nash Equilibrium of the game, i.e., the equilibrium bidding strategies of the buyers and what this equilibrium strategies imply in terms of revenue for the seller.
- Moral of the theory: incomplete information matters! Most situations we encounter in daily life feature incomplete information, an auction theory is a gateway to begin to understand them (from a theoretical perspective).
- The theory of auctions is quite elegant, but is not simple.
- The mathematical challenge together with the clear applicability makes the theory of auctions **central** to modern economics.

Definition 5.1

A **independent private values sealed-bid auction** is a tuple $(N, \{\mathcal{V}_i, F_i\}_{i \in N}, q, p)$ where:

- $N = \{1, 2, \dots, n\}$ is the set of buyers or bidders.
- $\mathcal{V}_i \subset \mathbb{R}$ is the set of **possible values** for buyer i for each $i \in N$. Denote $\mathcal{V} = \prod_{i \in N} \mathcal{V}_i$.
- For each buyer $i \in N$, F_i is the cumulative distribution function over her set of values $\mathcal{V}_i$.
- $q : [0, \infty)^n \rightarrow \Delta(N)$ is a function associating each vector of bids $b \in [0, \infty)^n$ with a distribution according to which the buyer who wins the object is identified.
- $p : N \times [0, \infty)^n \rightarrow \mathbb{R}^n$ is a function determining the payment each buyer pays, for each vector of bids $b \in [0, \infty)^n$, depending on which buyer $i_* \in N$ is the winner.

- The private value $v_i \in \mathcal{V}_i$ is chosen randomly according to F_i .
- Every buyer i learns her private value v_i , but not the private value of the other buyers.
- Every buyer i submits a bid $b_i \in [0, \infty)$, which depends on her private value v_i .
- The buyer(s) who win the auctioned object, i_* , is chosen according to the distribution $q(b_1, \dots, b_n) \in \Delta(N)$. The probability that buyer i wins the object is $Q_i(b_1, \dots, b_n)$.
- Every buyer i pays the sum $p_i(i_*; b_1, \dots, b_n)$.

The sealed-bid private values auction as a Bayesian Game

Note that the private value of buyer i is a random variable with CDF F_i . This random value will be denoted as V_i . A sealed-bid auction is a game with incomplete information, which can be described as follows:

- The set of players is the set of buyers $N = \{1, \dots, n\}$.
- Player i 's set of types is $\mathcal{V}_i$. Let $\mathcal{V} = \prod_{i \in N} \mathcal{V}_i$, that is the set of type vectors.
- The set of states of the world is $\mathcal{V}$.
- The prior distribution over the set of type vectors is a product distribution $F = F_1 \times F_2 \times \dots \times F_n$.
- The signalling functions are $\tau_i : \mathcal{V} \rightarrow \mathcal{V}_i$ are just the projections to the individual components: if the type vector is $v = (v_i)_{i \in N}$, then $\tau_i(v) = v_i$.
- For each type (value) vector $v \in \mathcal{V}$, buyer's i profit is given by:

$$u_i(b, v_i) = q_i(b)v_i - \sum_{i_* \in N} q_{i_*}(b)p_i(i_*; b)$$

What about open-bid auctions?

Remark: Open-bid auctions

A formal definition of an open-bid depends on the specifics of the method used to conduct the auction, which may be extremely complex. The tool required to do so would lead us to study **extensive form games**, and the appropriate solution concepts for extensive forms.

For example, in the English Auction, a buyer's decision on when to stop bidding depends:

- the private value of the bidder,
- the identities of the buyers that have already quit and those that still remain,
- the prices at which those that have already quit chose to stop bidding

Independent private values sealed-bid auction: payoffs

- Let $\beta_i : \mathcal{V}_i \rightarrow [0, \infty)$ be a pure-strategy for each player i . Let $\beta(v) = (\beta_1(v_1), \dots, \beta_n(v_n))$ be a strategy profile.
- The ex-ante expected payoff to player i is:

$$U_i(\beta) = \mathbb{E}[u_i(\beta(V))] = \int_{v \in \mathcal{V}} u_i(\beta_1(v_1), \dots, \beta_n(v_n)) dF(v)$$

- Let $\beta_{-i}(v_{-i}) = (\beta_j(v_j))_{j \neq i}$, i.e., the vector of pure strategies of all players except i .
- The interim expected payoff to player i when her type is $V_i = v_i$ is:

$$U_i(\beta; v_i) = \mathbb{E}[u_i(\beta_i(v_i), \beta_{-i}(V_{-i})) | V_i = v_i] = \int_{v_{-i} \in \mathcal{V}_{-i}} u_i(\beta_i(v_i), \beta_{-i}(v_{-i})) dF_{-i}(v_{-i})$$

where $\mathcal{V}_{-i} = \prod_{j \neq i} \mathcal{V}_j$ is the space of buyer values other than i , and $F_{-i} = \prod_{j \neq i} F_j$ is the cumulative joint distribution function of the random variable $V_{-i} = \prod_{i \in N} V_j$.

- Notice that the interim expected profit of buyer i when her value is v_i depends on β_i , buyer i 's strategy, only via $\beta_i(v_i)$, which is the bid of buyer i with private value v_i .
- Denote by $U_i(b_i, \beta_{-i}; v_i)$ the expected profit of buyer i with private value v_i when she submits a bid b_i and the other buyers use the profile β_{-i} .

Definition 5.2

A strategy profile β^* is an equilibrium (Bayesian-Nash eq.) if for every buyer $i \in N$, and every private value $v_i \in \mathcal{V}_i$:

$$U_i(\beta^*; v_i) \geq U_i(b_i, \beta_{-i}^*; v_i), \quad \forall b_i \in [0, \infty).$$

Equilibrium in the sealed-bid second price auction

- Let $N(b) = \{i \in N : b_i = \max_{j \in N} b_j\}$.
- The function $Q(b) \in \Delta(N)$ of the second-price auction is:

$$Q_i(b) = \begin{cases} 0 & i \notin N(b) \\ \frac{1}{|N(b)|} & i \in N(b) \end{cases}$$

- Let i_* denote the buyer who wins the auctioned object, then the payment rule $P(i_*; b)$ is:

$$P_i(i_*; b) = \begin{cases} 0 & i \neq i_* \\ \max_{j \neq i} b_j & i = i_* \end{cases}$$

- Notice that $u_i(b, v_i)$ becomes:

$$u_i(b, v_i) = \begin{cases} 0 & b_i < \max_{j \in N} b_j \\ \frac{v_i - \max_{j \neq i} b_j}{|N(b)|} & b_i = \max_{j \in N} b_j \end{cases}$$

Equilibria in the SPA and the English Auction with private values.

Theorem 5.3

In a sealed-bid second price auction, the strategy of buyer i in which he bids his private value (ex-post) weakly dominates her other strategies.

Theorem 5.4

In a sealed-bid second price auction, the strategy profile in which every buyer's bid equals her private value is an equilibrium.

Theorem 5.5

In an open-bid ascending auction (like the English auction), the strategy profile in which every buyer lowers her hand when the declared price equals her private value weakly dominates all his other strategies.

Theorem 5.6

In an open-bid ascending auction, the strategy profile in which every buyer lowers her hand when the declared price equals her private value is an equilibrium.

The symmetric model with independent private values

We will now study a class of models that satisfy the following assumptions:

- A1 Single object for sale and the object is indivisible.
- A2 The seller is willing to sell the object at any nonnegative price.
- A3 There are n buyers.
- A4 Private values and symmetric values: All buyers have the same set of possible private values $\mathcal{V}$. This set can be a closed bounded interval $[0, \bar{v}]$ or the set of nonnegative numbers $[0, \infty)$. Every buyer knows her private value for the object. The random values $V_1, V_2, \dots, V_n$ of the private values of the buyers are independent and identically distributed. Denote by F the common cumulative distribution function of the random variables V_i , $i = 1, \dots, n$. The support of this distribution is $\mathcal{V}$. The joint distribution is then $F^n = F \times \dots \times F$.
- A5 Continuity: For each i , the random variable V_i is continuous, and its density function, denoted by f_i , is continuous and positive.
- A6 Risk neutrality: All buyers are risk neutral, and therefore seek to maximize their expected profits.

A1–A6 are common knowledge among the buyers. Any auction satisfying A1–A6 is called a **symmetric auction with independent private values (SIPV)**.

Definition 5.7

In a symmetric auction with independent private values, an equilibrium profile $(\beta_1^*, \beta_2^*, \dots, \beta_n^*)$ is called a **symmetric equilibrium** if $\beta_i^* = \beta_j^*$ for all $i, j \leq n$; that is, all buyers implement the same strategy.

When the profile $\beta^* = (\beta_i^*)_{i \in N}$ is a symmetric equilibrium, we *abuse notation* and denote the common strategy also as β^* , i.e., $\beta^* = \beta_i^*$ for every $i \in N$. Such a strategy is called a **symmetric equilibrium strategy**.

Definition 5.8

A symmetric equilibrium strategy β^* is monotonically increasing if the higher the private value, the higher the buyer's bid:

$$v < v' \Rightarrow \beta^*(v) < \beta^*(v') \quad \forall v, v' \in \mathcal{V}.$$

Example: Two bidders with i.i.d Uniform valuations

Example 5.9

Suppose that V_1 and V_2 are independent and identically distributed (i.i.d.) Uniform random variables over the interval $[0, 1]$, i.e., $V_1, V_2 \sim U[0, 1]$. In this example, we will show that in a sealed-bid first-price auction, the following strategy is a symmetric equilibrium:

$$\beta_i^*(v_i) = \frac{v_i}{2} \quad i = 1, 2.$$

Heuristic derivation of the symmetric eq. strategy with n bidders in the SIPV First-Price Auction.

Suppose that bidders $j \neq 1$ follow the symmetric, monotonically increasing, and differentiable equilibrium strategy β . Suppose that bidder 1 has value $V_1 = v_1$ and bids b .

Theorem 5.10

Let $Y = \max_{j \neq i} \{V_j\}$ be the random variable whose value equals the highest private value of all buyers j other than i , and let G and g denote respectively the CDF and PDF of Y . In a symmetric first-price auction with independent private values the following strategy defines a monotonically increasing, continuous, differentiable, symmetric equilibrium:

$$\beta(v) = E[Y | Y \leq v], \quad \forall v \in \mathcal{V} \setminus \{0\}$$

and $\beta(0) = 0$.

Proof.

Step 1: $\beta(v)$ is monotonically increasing.

Proof (Cont.)

Step 2: $\beta(v)$ is continuous.

Step 3: $\beta(v)$ is differentiable.

Proof (Cont.)

Step 4: $\beta(v)$ is a symmetric equilibrium.



Theorem 5.11

In the symmetric equilibrium given by Theorem 5.10, the expected payment that a buyer with private value v makes for the object is $G_Y(v) \times E[Y | Y \leq v]$ in both sealed-bid first-price and second-price auctions.

Proof.



Theorem 5.12

In a symmetric sealed-bid auction with independent private values, at the symmetric equilibrium, the expected revenue of a seller in both sealed-bid first-price and second-price auctions, is

$$\pi = n \int_{v \in \mathcal{V}} G_Y(v) \mathbb{E}[Y | Y \leq v] f(v) dv$$

Proof.

